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$$\begin{aligned}
 & \lim_{x \rightarrow a} \frac{x\sqrt{x} - a\sqrt{a}}{x - a} \\
 &= \lim_{x \rightarrow a} \frac{(x\sqrt{x} - a\sqrt{a}) \cdot (x\sqrt{x} + a\sqrt{a})}{(x - a) \cdot (x\sqrt{x} + a\sqrt{a})} \\
 &= \lim_{x \rightarrow a} \frac{x^3 - a^3}{(x - a) \cdot (x\sqrt{x} + a\sqrt{a})} \\
 & (\Rightarrow a^3 - b^3 = (a - b)(a^2 + ab + b^2)) \\
 &= \lim_{x \rightarrow a} \frac{\cancel{(x - a)} \cdot (x^2 + ax + a^2)}{\cancel{(x - a)} \cdot (x\sqrt{x} + a\sqrt{a})} \\
 &= \frac{3a^2}{2a\sqrt{a}} \\
 &= \frac{3a\sqrt{a}}{2a} \\
 &= \frac{3\sqrt{a}}{2}
 \end{aligned}$$

References